

Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Micro-Project

PHASE 5 | PROJECT DEVELOPMENT PHASE

5. Project Development Phase

This phase implements the design defined in Phase 3, following the milestone sequence from Phase 4. Each step below reflects the actual configuration performed on the ServiceNow instance, with output screens captured directly from the build.

5.1 Milestone 1 — User Creation

1. Navigate to **User Administration** → **Users** and click **New**.
2. Set User ID: `EEEUser`, First name: `EEE`, Last name: `User`, Email: `eeuser@gmail.com`.
3. Click **Update** to save the record.

The screenshot shows the 'New User' form in ServiceNow. The form is titled 'User Administration' and 'EEE User'. The 'User ID' field is set to 'EEEUser', 'First name' is 'EEE', and 'Last name' is 'User'. The 'Email' field is set to 'eeuser@gmail.com'. The 'Identity type' is 'Human', 'Language' is '-- None --', 'Calendar integration' is 'Outlook', 'Time zone' is 'System (Asia/Kolkata)', and 'Date format' is 'System (yyyy-MM-dd)'. The 'Active' checkbox is checked. The 'Update' button is highlighted with a red box.

Output screen — EEE User record created with User ID, name, and email populated.

5.2 Role Creation

1. Navigate to **User Administration** → **Roles** and click **New**.
2. Enter Name `bb1` and a short description, then click **Submit**.
3. Repeat for `bb2`, `bb3`, and `bb4`.
4. Assign all four roles to the EEE User record.

The screenshot shows the 'New Role' form in ServiceNow. The form is titled 'Role Administration' and 'BB1'. The 'Name' field is set to 'BB1' and the 'Application' is set to 'Global'. The 'Update' button is highlighted with a red box.

Output screen — Custom role BB1 created (the same procedure was repeated for BB2, BB3, BB4).

5.3 Table Creation

1. In the Application Navigator, search **System Definition** → **Tables** and click **New**.
2. Label: `Institution Details`, Name: `u_institution_details`, Extends: `false`. Click **Submit**.

The screenshot shows the 'New Table' form in ServiceNow. The form is titled 'Table Administration' and 'Institution Details'. The 'Label' field is set to 'Institution Details' and the 'Name' field is set to 'u_institution_details'. The 'Update' button is highlighted with a red box.

Output screen — Institution Details table created (`u_institution_details`).

5.4 Field Creation

The following fields were added to `u_institution_details`:

Field	Type
Student Roll Number	Auto Number

Student Name	Reference - User
Faculty Name	Reference - User
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi-line String

Column label	Type	Reference	Max length	Default value
Updates	Integer	(empty)	40	
Created by	String	(empty)	40	
Student Roll Number	String	(empty)	40	javascript:global.getNextObj()NumberPadded();
Description	String	(empty)	1,000	
Updated by	String	(empty)	40	
Phone Number	String	(empty)	40	
Sys ID	Sys ID (GUID)	(empty)	32	
Created	Date/Time	(empty)	40	
Faculty name	Reference	User	32	
Branch	String	(empty)	40	
Updated	Date/Time	(empty)	40	
Student Name	Reference	User	32	
Email	String	(empty)	40	
Insert a new row...				

Output screen — Dictionary entries for the Institution Details table showing all created fields.

5.5 Seed Records

Multiple records were created covering all three branch values so that the ACL's filtering effect could be observed clearly during testing.

Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

Output screen — Institution Details records seeded with ECE, EEE, and CSE branch values.

5.6 Elevating to security_admin

Before any Access Control record can be created or edited, the security_admin role must be elevated from the user profile menu.

Elevate role

Close dialog

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel

Update

Output screen — security_admin role elevated to allow ACL modification.

5.7 Read ACL — Script & Data Condition

1. Navigate to **System Security → Access Control (ACL)** and click **New**.
2. Type: record, Operation: read, Name: u_institution_details, Active: true, Advanced: true.
3. Requires role: bb1. Data Condition: Branch is EEE.
4. Advanced script:

```
(function () {  
  // Allow admin users full access  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  // Allow only EEE branch users to see EEE records  
  if (gs.hasRole('bb1')) {  
    return true;  
  }  
  // Deny access for all others  
  return false;  
})();
```

Data Condition

Condition No. of records matching the condition: 2

Add Filter Condition

Add OR Clause

Branch is EEE

AND

OR

×

Advanced Condition

Controlled by References



Script Turn on ECMAScript 2021 (ES12) mode



```
1  
2 (function() {  
3   if (gs.hasRole('bb1')) {  
4     return true;  
5   } else {  
6     return false;  
7   }  
8 })();
```

Output screen — Read ACL Data Condition (Branch is EEE) and Advanced Condition script.

Access ControlsUpdatedSearch

All

Name	Decision Type	Operation	Type	Active	Updated by	Updated
sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago
sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04
sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04

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Output screen — Access Control list confirming the new Read ACL against u_institution_details.

5.8 Create ACL

Type: record, Operation: create, Name: u_institution_details, Active: true. Requires role: bb2. No Data Condition needed.

Access Controlu_institution_details

Type: record

Operation: create

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

Name: Institution Details [u_institution_details]

Description:

Applies To: No. of records matching the condition: 6

Application: Global

Active: ☒

Advanced: ☐

-- None --

-- choose field -- -- oper -- -- value --

UpdateDelete

Output screen — Create ACL configured for u_institution_details.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role

Role

BB2

Insert a new row...

Security Attribute Condition

Output screen — Requires role list showing BB2 attached to the Create ACL.

5.9 Write ACL

Type: record, Operation: write, Name: u_institution_details, Active: true. Requires role: bb3. No Data Condition needed.

Access Control
u_institution_details

* Type: record

* Operation: write

Decision Type: Allow If

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: institution Details [u_institution_details] -- None --

Description: </>

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Output screen — Write ACL configured for u_institution_details.

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role	
Role	
BB3	

Insert a new row...

Output screen — Requires role list showing BB3 attached to the Write ACL.

5.10 Delete ACL

Type: record, Operation: delete, Name: u_institution_details, Active: true. Requires role: bb4. No Data Condition needed.

Access Control
New record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: delete

Decision Type: Allow If

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: institution Details [u_institution_details] -- None --

Description: </>

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Output screen — Delete ACL record created for u_institution_details.

Requires role	
Role	
BB4	

Insert a new row...

Security Attribute Condition

Output screen — Requires role list showing BB4 attached to the Delete ACL.

All four CRUD-level, script-controlled ACLs (Read, Create, Write, Delete) were successfully implemented against u_institution_details, each requiring its respective custom role (bb1-bb4) and, for Read, an additional Branch = EEE data condition. Development is now ready for structured testing (Phase 6).